

The Future Drought Fund's Science to Practice Forum brings together farmers, researchers, government, industry and the community to explore the innovative tools and practices helping our farmers and regions prepare for future drought.

The online forum also showcases the work of the Future Drought Fund's [8 Drought Resilience Adoption and Innovation Hubs](#) located around the country.

Watch the Future Drought Fund's 2023 Science to Practice Forum below and see [all the 2023 forum video content on Vimeo](#).

Introduction

The Future Drought Fund's promotional video for the Science to Practice Forum featuring highlights from the 2023 Science to Practice Forum.

Transcript

[Video begins]

Speaker 1:

Speaker 2:

We'll hear directly from Australian farmers and communities who are preparing for future drought by utilising innovative tools and practices.

Speaker 3:

Ever since I left school, Australia's been talking about drought or in major droughts. It's always going to happen, so as long as we can prepare for the next one.

Speaker 4:

The Future Drought Fund has been pivotal in enabling collaboration across research development, extension, commercialisation, and adoption.

Speaker 5:

Part of our role is to be able to share the learnings and actually getting people boots on the ground, you know, getting their fingers in the dirt and seeing the difference that it's making out in the paddock.

Speaker 6:

It's all about putting in place foundations for farmers and communities to draw upon when the next drought hits so they can better manage through it.

Speaker 7:

And that's why this program I'm so excited about and the funding from the federal government because it's actually doing the preparation work, tackling it when we've got a chance to think about it rather than actually under the pressure of trying to respond.

Speaker 8:

The Future Drought Fund, I think, is actually really fundamentally important to the future of our Australian regional communities.

Speaker 9:

We can actually come to have some control over climate change.

Watch the 2023 Forum

00:00:00- Welcome – Pip Courtney, Forum Host

00:01:47 - Welcome to Ngunnawal Country – Tyronne Bell, Ngunnawal Traditional Owner

00:04:57 - Welcome to the 2023 Science to Practice Forum – Senator the Hon Murray Watt, Minister for Agriculture, Fisheries and Forestry

00:07:33 - FDF update – First Assistant Secretary Mel Brown and Acting Assistant Secretary Laura Smith, Department of Agriculture, Fisheries and Forestry

00:25:15 - FDF Consultative Committee update – Caroline Welsh, FDF Consultative Committee

[Day 1 - Welcome to the Science to Practice Forum](#) from [DAFF Future Drought Fund](#) on [Vimeo](#).

Session 1 – Setting the scene for sustainable agriculture: 6 June 2023

Presentation: Climate outlook and long-range forecasts - Felicity Gamble, Senior Climatologist, Bureau of Meteorology (BoM)

Video: Optimal management of soils and available water in NSW - FDF Drought Resilient Soils and Landscapes project

Feature speaker: Joshua Gilbert, Worimi producer and First Nations farming advocate

[Day 1 Session 1 - Setting the scene for sustainable agriculture](#) from [DAFF Future Drought Fund](#) on [Vimeo](#).

Session 2 – Seeds and soil health: Innovating at scale: 6 June 2023

00:00:00- Video: Wattle seed enterprise in WA – SW WA Hub, Noongar LEG

00:03:31 - Presentation: FDF Drought Resilience Commercialisation Initiative – Blake Zur, Acting Principal Director, Farm Resilience Branch, Future Drought Fund

00:13:19 - Presentation: Modern mulch – Dr Stuart Gordon, CSIRO & Richard Fennessy, DPIRD

00:34:58 - Video: Soil Moisture Monitoring, WA – SW WA Hub, DPIRD

00:40:14 - Presentation: Soil moisture monitoring to improve irrigation – Neil Lantzke, Senior Research Scientist, DPIRD

00:52:13 - Presentation: Data driven drought resilience in Vic – Bronwyn Bant, Regional Agriculture Landcare Facilitator (RALF), Wimmera CMA

[Day 1 Session 2 – Seeds and soil health: Innovating at scale from DAFF Future Drought Fund on Vimeo.](#)

Session 3 & 4 – Global connections for drought resilience & Where to next: 6 June 2023

Global connections for drought resilience

00:00:00- Video: Agscent technology in action in Qld – TNQ Hub

00:06:32- Panel: Nuffield Scholars on drought resilience – Dr Ellen Litchfield, Andrew Rolfe, Jesse Moody

00:56:15- Video: Filling feed gaps with saltbush in WA – FDF Drought Resilient Soils and Landscapes grant project delivered by CSIRO

00:59:23- Presentation: I'll drink to that: Drought resilient wine varieties – Dr Alexander Copper, Alexander Willem Wines

01:15:24- Video: Building drought resilience through soil health trial in Qld – FDF NRM Drought Resilience Program - Grants project

Where to next?

01:20:05- Presentation: Next phase of the FDF – Louise Palfreyman, Director, Farm Resilience Division, DAFF

01:32:03- Day 1 closing remarks – Pip Courtney, Forum Host

[Day 1 Sessions 3 and 4 - Global connections for drought resilience & Where to next? from DAFF Future Drought Fund on Vimeo](#)

Welcome and Session 1 – Putting down roots: 7 June 2023

Welcome

00:00:00 - Welcome – Pip Courtney, Forum Host

00:01:46 - Welcome to Ngunnawal Country – Tyronne Bell, Ngunnawal Traditional Owner

00:04:08 - Welcome to Day 2 of the 2023 Science to Practice Forum – Secretary Andrew Metcalfe AO, Department of Agriculture, Fisheries and Forestry

Putting down roots

00:12:07 - Feature speaker – Michael Taylor: 2022 Australian Farmer of the Year and sixth generation farmer

00:51:44 - Video: Regional Soil Coordinators Community of Practice field trip in Tasmania

00:57:54 - Video: Rehydrating landscapes in the Northern Territory

[Day 2 Welcome and Session 1 - Putting down roots](#) from [DAFF Future Drought Fund](#) on [Vimeo](#)

Session 2 – Planning for profitability and productivity in a changing climate: 7 June 2023

00:00:00 - Presentation: Seasonal forecasting for drought resilience – Emeritus Prof. Roger Stone, UniSQ (SQNNSW Hub).

00:05:31 - Presentation: Insights from a Senior Rural Financial Counsellor – Roly Chugg, Senior Rural Financial Counsellor, Rural Business Tasmania/Rural Financial Counselling Service (RFCS, DAFF).

00:20:18 - Video: Farming Family Reboot (SQNNSW Hub).

00:29:01 - Presentation: Improving the odds for farm profitability – Kerry Battersby, Project Manager, Queensland Farmers Federation (QFF).

00:44:54 - Presentation: The importance of a baseline carbon audit – Glenn Briggs, National Business Development Manager, Aglytica Pty Ltd.

00:57:18 - Presentation: Growing success: The benefits of Natural Capital Accounting – Dr Daniel Mendham, Senior Research Scientist, CSIRO.

[Day 2 Session 2 - Planning for profitability and productivity in a changing climate](#) from [DAFF Future Drought Fund](#) on [Vimeo](#)

[Video transcript: 2023 Science to Practice Forum Day 2 Session 2 \(DOCX 254 KB\)](#)

Session 3 – Farming through drought: Lived experiences and lessons learned: 7 June 2023

00:00:00 - Presentation: Sowing the Gap: Filling feed gaps to better manage stock – Jaymie Dawes, Program Manager Community and Partnerships, East Gippsland Catchment Management Authority (EGCMA)

00:11:47 - Video: Sowing the Gap – Chris Nixon farm study, Vic

00:14:51 - Video: Sowing the Gap – Scott Langley farm study, Vic

00:18:40 - Panel: Young Agfluencers – Nick Robinson, Mark Merrett, Edwina Robertson

[Day 2 Session 3 - Farming through drought: Lived experiences and lessons learned from DAFF Future Drought Fund on Vimeo](#)

Session 4 – Tools for change: 7 June 2023

00:00:00 - Video: Remote water monitoring in SA (SA Hub)

00:05:09 - Presentation: Right tool for the right job – Kara-Glenn Worth, Digital Ag Innovation Consultant, TNQ Hub

00:19:01 - Presentation: Climate Services for Agriculture (CSA) – Stephanie Dickson, Customer Engagement Coordinator, Bureau of Meteorology (BoM)

00:38:58 - Video: Containment feeding in Vic (Vic Hub)

00:45:46 - Presentation: Transferrable skills in resilience – Kristie Lisle, Beef producer, Lisle Holdings Pty Ltd.

00:58:46 - Video: Farm Business Resilience Program (FBRP) – Cypress Springs ACT

01:02:43 - Video: Climate Friendly: Carbon farming for drought resilience

01:11:36 - Day 2 closing remarks – Pip Courtney, Forum Host

[Day 2 Session 4 - Tools for change from DAFF Future Drought Fund on Vimeo](#)

Welcome and Session 1 – Relationships for resilience: 8 June 2023

Welcome

00:00:00 – Welcome – Pip Courtney, Forum Host

00:01:46 - Welcome to Nggunawal Country – Tyronne Bell, Nggunawal Traditional Owner

00:03:47 - Welcome to Day 3 of the 2023 Science to Practice Forum – Deputy Secretary Rosemary Deininger, Department of Agriculture, Fisheries and Forestry Session 1: Relationships for resilience

00:09:33 - Reflections on Days 1 & 2 – Host Pip Courtney

00:13:55 - Feature speaker – Julia Spicer OAM, Queensland's Chief Entrepreneur

00:35:36 - Presentation: Foundation for Rural & Regional Renewal (FRRR) – Natalie Egleton, CEO, FRRR

01:07:33 - Video: Outback Academy (FRRR)

01:13:28 - Presentation: Australian Rural Leadership Foundation (ARLF)

01:38:47 - Video: Regional Drought Resilience Planning (RDRP) in the Murraylands and Riverland LGA, SA

01:45:37 - Presentation: Murraylands and Riverland Plan - a Regional Drought Resilience Plan – Ben Fee, Chief Executive and Analise Gates, Marketing and Strategic Reporting Manager, Regional Development Australia Murraylands and Riverland

01:55:02 - Presentation: Using drought resilience to inspire innovation in ag finance – Veronica Terry, Regional Economic Development Coordinator, Regional Development Australia Tasmania

02:08:31 - Presentation: Using the Knowledge Broker network for disruptive design – Dale Stringer, Chief Knowledge Broker, SNSW Hub

02:22:49 - Video: Regional Drought Resilience Planning in Wimmera, Vic

[Day 3 Welcome and Session 1 - Relationships for resilience from DAFF Future Drought Fund on Vimeo](#)

Session 2 – Engaging First Nations farmers: 8 June 2023

00:00:00 - Presentation: First Nations stakeholder engagement – Dr Raelene Ward, Knowledge Broker First Nations Engagement, SQNNSW Hub

00:17:18 - Presentation: Sustainable Aboriginal and Torres Strait Islander Enterprise (SATSIE) program – Allan Dale, Professor of Tropical Regional Development, James Cook University, Program Lead SATSIE, TNQ Hub

00:36:58 - Presentation: Building Human Capacity – Jane Oorschot, Program Lead Building Human Capacity, TNQ Hub

[Day 3 Session 2 - Engaging First Nations farmers from DAFF Future Drought Fund on Vimeo](#)

Session 3 – Overcoming barriers to change: 8 June 2023

00:00:00 - Presentation: Choosing resilience: surveying farmers across Australia – Dr Hanabeth Luke, Soil CRC Survey Project Leader, Southern Cross University

00:12:09 - Presentation: Improving drought resilience through behavioural science – Sam Moore, Senior Behavioural Scientist, Evidn and Daniel Rea, Partnership Coordinator, Fitzroy Basin Association (FBA)

[Day 3 Session 3 - Overcoming barriers to change from DAFF Future Drought Fund on Vimeo](#)

Session 4 – Mental health heroes: 8 June 2023

00:00:00 - Presentation: Writing a new chapter with My Drought Story – Kate Arndell, Rural Adversity Mental Health Program Coordinator, NSW Health

00:16:36 - Panel: For the love of farming - the importance of talking when times are tough – Andrew Daley, Brad Millsteed, Big Farmer Andy

00:56:34 - Presentation: Vocal Locals find their voice – Kate Gunn, Associate Professor, ifarmwell at University of South Australia
01:12:59 Forum closing remarks: Host Pip Courtney

[Day 3 Session 4 - Mental health heroes](#) from [DAFF Future Drought Fund](#) on [Vimeo](#)

Watch the 2022 Forum

Part 1 Future Drought Fund Science to Practice Forum: 7 June 2022

00:00:49 - Welcome to Country – Nin Jannette Phillips

00:03:13 - Our progress towards drought resilience – Brent Finlay & Caroline Welsh

00:14:54 - How the Future Drought Fund supports Australians in preparing for drought – Louise Palfreyman & Kimberley Shrives

Harnessing innovation Part 1

00:35:10 - SWWA snapshot video

00:40:26 - SWWA showcase: Innovative water supply options for the future – Richard George & Nik Callow

01:00:58 - Disruptive agricultural innovation: Resilient production systems for a changing climate – Cal Archibald

01:21:02 - Walking the talk of drought innovation – Dr Alison Southwell

01:36:46 - SQNNSW snapshot video

01:41:03 - SQNNSW showcase: Using weather data to predict risks & opportunities for livestock farmers with Ag360

Part 2 Future Drought Fund Science to Practice Forum: 7 June 2022

Harnessing innovation Part 2

00:01:24 - TNQ snapshot video

00:06:50 - TNQ showcase: Commercialising farmers' ideas through the TNQ Hatch program – Darryl Lyons

00:20:23 - Protecting Indigenous Cultural and Intellectual Property while innovating in agriculture – Laura Melrose & Joel Murgha

00:39:28 - Sharing drought resilient animal production knowledge via the Hubs – Sarah Strachan

00:54:19 - Cautionary tales from the WA drought reform measures pilot program (2010-11) and other experiences (Main Stage) – Blake Zur, John Noonan, Dr Christine Storer, Rebecca Heath & Dr Roy Murray-Prior

01:24:32 - Drought Resilience Mission: Putting CSIRO science into practice – Dr Graham Bonnett

Part 3 Future Drought Fund Science to Practice Forum: 7 June 2022

Better land management

00:01:11 - NWANT snapshot video

00:05:39 - NWANT showcase: Applying best practice pastoral land management to Aboriginal lands for future drought – Multiple people

00:19:54 - The Noongar Boodjar Rangers saving seed to heal land and build drought resilience – Dr Karl O'Callaghan & Jermaine Davis Sr

00:39:29 - Reading the landscape to identify disturbances and fortify farming systems – Glenn Landsberg:

00:53:23 - PANEL: How do you build drought resilience in organic production systems? – Niki Ford, Melinee Leather, Nathan Moore & David Keens

01:31:40 - SA snapshot video

01:37:02 - SA showcase: Supporting farmers along Goyder's Line – Taryn Mangelsdorf

Part 1 Future Drought Fund Science to Practice Forum: 8 June 2022

00:00:27 - Welcome – Andrew Metcalfe AO

Soils in focus

00:07:58 - New resources for managing soils for drought – Dr Michael Crawford

00:28:25 - PANEL: How do we best manage soils for drought resilience? – Troy Clarkson, Dr Michael Crawford, Naveeda Majid & Belinda Eastough

Improved knowledge

01:14:02 - From the Great Barrier Reef to the land: carbon markets and natural capital accounting for drought resilience – Mike Berwick

01:31:48 - Helping farmers find the sweet spot between productivity and natural resource management – Dr Sue Ogilvy

01:46:28 - Smart Farms: A Grower Group's efforts to share data and increase productivity – Nathan Dovey & Phillip Honey

01:59:46 - The essentials when engaging with Indigenous communities – Halena Scanlon & Brian Arley

Part 2 Future Drought Fund Science to Practice Forum: 8 June 2022

Better risk management

00:01:22 - TAS snapshot video

00:06:20 - TAS showcase: Supporting farmers to improve their financial stability through business coaching – Roly Chugg

00:28:14 - The agricultural, research and insurance industries improving income stability for crop farmers together – Jo Sheppard & Kerry Battersby

00:48:42 - Systemic risk management in farming – geographic distribution and diversity – Dr Thong Nguyen-Huy:

00:56:55 - Supporting farmers and regions to better plan for and manage drought risk- Alex Hicks & Henry Moxham

Part 3 Future Drought Fund Science to Practice Forum: 8 June 2022

More resilient communities

00:00:45 - SNSW snapshot video

00:05:56 - SNSW showcase: Mindset shifts and their power in collaboration – Jo Eady

00:26:48 - An update in growing farmers' wellbeing online – Dr Kate Gunn

00:40:04 - PANEL: How do communities build and maintain their drought resilience? – Saleena Ham, Natalie Eggleton, Dr Jennifer Luke & Simon Millcock

01:23:57 - VIC snapshot video

01:28:56 - VIC showcase: Using creativity and imagination to strengthen community resilience to future drought – Dr Suzanne Fraser

01:46:43 - Strengthening connections between young farmers and their information and support networks – Amanda Scott & Naomi Scholz

02:03:38 - Closing – Lyn O'Connell

Watch the 2021 Forum

Part 1 Science to Practice Forum: Building the Foundations of Drought Resilience 29 June to 1 July 2021

- 00:00:40 – Welcome to Country: Warren Daley, Ngunnawal Elder
- 00:05:12 – Opening Address: Mr Brent Finlay, Chair of the FDF Consultative Committee
- 00:10:34 – Vision for the Drought Resilience Hubs to inspire agricultural innovation: Andrew Metcalfe AO, Secretary, Department of Agriculture, Water and the Environment
- 00:15:58 – The Future Drought Fund: Brent Finlay, Chair of the FDF Consultative Committee and Louise Palfreyman, A/Assistant Secretary, Department of Agriculture, Water and the Environment
- 00:50:24 – Innovation driving science and exploration: Lloyd Symons, Polar Technology Manager, Australian Antarctic Division, Department of Agriculture, Water and the Environment
- 01:25:30 – A vision for co-designed drought research: Dr Graham Bonnett, Interim Leader Drought Resilience Mission, CSIRO

Part 2 Science to Practice Forum: Building the Foundations of Drought Resilience 29 June to 1 July 2021

- 00:00:00 – Hubs in focus:
- 00:04:51 – Tropical North Queensland Hub - James Cook University
- 00:39:57 – Southern Queensland and Northern New South Wales Hub - University of Southern Queensland
- 01:07:09 – Southern New South Wales Hub - Charles Sturt University
- 01:39:37 – The National Drought Resilience Research and Adoption Investment Plan: Michele Akeroyd, Director, Department of Agriculture, Water and the Environment and Alluvium Consulting.

Part 3 Science to Practice Forum: Building the Foundations of Drought Resilience 29 June to 1 July 2021

- 00:00:46 – Networks to Build Drought Resilience & Drought Resilience Leaders: Nina O'Brien, Disaster Resilience & Recovery Lead, Foundation for Rural & Regional Renewal, and Philippa Woodhill, Partnerships Director, Australian Rural Leadership Foundation
- 00:42:19 – Getting Young Farmers off their Tractors: Alexandra Hicks, Young Farmer Business Program, New South Wales Department of Primary Industries
- 00:55:26 – Community engagement strategies to strengthen the preparedness of First Nations People through the adoption of drought resilient processes: Dr Raelene Ward, School of Nursing, University of Southern Queensland
- 01:15:12 – Drought, wellbeing and adaptive capacity in rural and regional Australia: Emma Austin and Associate Professor Anthony Kiem, School of Environmental and Life Sciences, University of Newcastle.

Part 1 Science to Practice Forum: Building the Foundations of Drought Resilience 29 June to 1 July 2021

- 00:00:00 – Welcome: Kerren Crosthwaite, First Assistant Secretary, Department of Agriculture, Water and the Environment
Hubs in focus:
- 00:11:43 – Tasmania Hub - The University of Tasmania
- 00:36:52 – Southwest Western Australia Hub - Grower Group Alliance
- 01:13:06 – Northern Western Australia and Northern Territory Hub - Charles Darwin University
- 01:50:38 – Jindaola - An Aboriginal Way of Embedding Knowledges and Perspectives: Professor Pascal Perez, Director of the SMART Infrastructure Facility, University of Wollongong and Jade Kennedy, Academic Developer and lecturer in Indigenous Knowledges, University of Wollongong
- 02:20:13 – Droughtmasters in the Southern Rangelands: Debbie and Ashley Dowden, Challa Station
- 02:33:04 – Climate Services for Agriculture and the Drought Resilience Self-Assessment Tool: CSIRO and the Bureau of Meteorology

Part 2 Science to Practice Forum: Building the Foundations of Drought Resilience 29 June to 1 July 2021

- Panel: What does a drought resilient future look like?
Dr Liz Petersen, FDF Consultative Committee, Principal Applied Economist and Director of Advanced Choice Economics, and Adjunct Senior Lecturer at the University of Western Australia
Enterprise Professor Bruce Pascoe, Indigenous Agriculture, University of Melbourne, and published author
Dr Kate Andrews, Executive Officer for NRM Regions Australia, and Senior Lecturer at the Fenner School of Environment and Society, Australian National University.

Part 3 Science to Practice Forum: Building the Foundations of Drought Resilience 29 June to 1 July 2021

- 00:00:00 – Northern Australia Climate Program: Dr Chelsea Jarvis, Centre for Applied Climate Sciences, University of Southern Queensland
- 00:09:15 – Demonstration of GrowAg, a tool developed by AgriFutures: Arianna Sippel, Senior Manager, GrowAg
- 00:24:29 – Harnessing biodiversity to build resilience and sustainability in diverse agricultural systems: Professor Geoff Gurr, School of Agricultural & Wine Sciences, Charles Sturt University
- 00:35:15 – Developing a framework for tactical decision making to address feed deficits: Dr Warwick Badgery, Rangelands & Tropical Pastures Research Leader, New South Wales Department of Primary Industries
- 00:41:24 – Building farmers' ability to manage effectively during drought: the www.ifarmwell.com.au approach: Dr Kate Gunn, Department of Rural Health, University of South Australia

- 00:53:40 – Queensland’s Drought and Climate Adaptation Program (DCAP): Dr Neil Cliffe, Program Manager of the Drought and Climate Adaptation Program, Queensland Department of Agriculture and Fisheries
- 00:59:41 – The Derwent Pasture Network, Tasmania: a dryland grazing community of practice: The Derwent Catchment Project
- 01:02:40 – Resilient Together: Lorraine Gordon, Director of Strategic Projects, Farming Together and the Regenerative Agriculture Alliance, Southern Cross University

Part 1 Science to Practice Forum: Building the Foundations of Drought Resilience 29 June to 1 July 2021

- 00:01:25 – Welcome: Joanna Stanion, First Assistant Secretary, Department of Agriculture, Water and the Environment
 - 00:08:58 – National Soil Strategy: Troy Clarkson, Assistant Director, Department of Agriculture, Water and the Environment
 - 00:34:47 – Agriculture Stewardship Package: Ryan Wilson, Department of Agriculture, Water and the Environment
 - 00:56:31 – Agricultural Innovation Agenda: Christine Mulhearn, Director, Department of Agriculture, Water and the Environment
- Science into practice: Anwen Lovett, Climate Research Strategy for Primary Industries
- 01:30:50 – Almond irrigation best practice management: Mark Skewes, Research Scientist, South Australian Research and Development Institute
 - 01:52:54 – Climate Smart Horticulture: Jessica Fearnley, Development Officer, New South Wales Department of Primary Industries, and Dr Allen Benter, Climate Applications and Digital Agriculture, New South Wales Department of Primary Industries
 - 02:23:37 – Rural Research and Development Corporation’s Climate Initiative: Matt Barwick, Research Portfolio Manager, Fisheries Research and Development Corporation
 - 02:45:00 – Agricultural Innovation Australia: Michelle Ford, Acting CEO, Agricultural Innovation Australia

Part 2 Science to Practice Forum: Building the Foundations of Drought Resilience 29 June to 1 July 2021

Hubs in focus

- 00:07:19 – Victoria Hub - The University of Melbourne
- 00:41:04 – South Australia Hub - The University of Adelaide
- 01:04:15 – The National Drought Resilience Research and Adoption Investment Plan: Reflections on priorities and innovations arising at the Forum: Alluvium Consulting
- 01:41:47 – Closing remarks: Lyn O’Connell, Deputy Secretary, Water, Climate Adaptation, Natural Disasters and Antarctic Group, Department of Agriculture, Water and the Environment